

Peiang (Ignatius) Miao

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RESEARCH INTERESTS

Multimodal agents, VLM, evidence-grounded reasoning, efficient and cost-aware inference, physical-world visual intelligence, digital dentistry, and clinical decision-support AI.

EDUCATION

The Hong Kong Polytechnic University

MSc, Metaverse Technology --- GPA 3.52/4.3, Credit

Department of Computing

Sep 2025 — Oct 2026

University College Dublin (joint program with Beijing University of Technology / BDIC)

B.Sc., Software Engineering---GPA 3.44/4.20, Upper Second Class Honours (parallel B.Eng.)

UCD China

Sep 2021 — Jun 2025

SELECTED PUBLICATIONS AND MANUSCRIPTS

Published

- H. Li, D. Li†, J. Li, **P. Miao**, Z. Zhong. "Reliability-Calibrated Mission-Window Assessment for Bearing Maintenance Decisions Under Operating-Condition Shift." *The Journal of Quality in Maintenance Engineering*. DOI: [10.1108/JQME-06-2026-0133](https://doi.org/10.1108/JQME-06-2026-0133).
- P. Miao**, S. Li, K. Cai, Y. Tian, R. Das, I. R. Nijhum, and S. Dev†. "EmoSense: AFRN-RAG Framework for FER-Based Facial Emotion Analysis and Mental Health Support." *33rd International Conference on Artificial Intelligence and Cognitive Science (AICS'25), Dublin*. DOI: [10.5281/zenodo.19141746](https://doi.org/10.5281/zenodo.19141746).

Under Review

- X. Wei*, **P. Miao***, et al. "[SpaGVAE: A Variational Graph Autoencoder with a Structured Spatial Prior for Robust Spatial Domain Identification](#)." Under review.
- X. Wei*, Y. Wang*, **P. Miao***, S. Mo, Y. Dong, R. Chen, H. Li, H. Lai†, and S. Qian†. "[Bio-CPD: A Distributional Divergence-based Framework for Single-cell Trajectory Analysis](#)." Under review.
- P. Miao***, H. Wang*, D. Li†, J. Li, and P. H. F. Ng. "Selective Vision-Language Inspection with Physics-Grounded Reasoning for Wafer Defect Recognition." Under review.
- R. Das*, **P. Miao***, and S. Dev†. "SS-VLM: A Spectral-Symbolic Vision-Language Framework for Auditable Facial Expression Recognition." Under review.
- X. Wei*, **P. Miao***, D. Li†, X. Fu, M. Liu, X. Lv, X. Wu, Z. Cai, Z. He, W. Yin, H. Lai†, and J. Shi†. "COMPASS: A Multi-Agent Framework for Clinical Research Protocol Development." Under review.
- P. Miao***, X. Wei*, D. Li†, Y. Jiang, L. Hu, K. Li. "RA-REF: Risk-Calibrated Relation-Evidence Fusion for Zero-Composite Industrial Defect Recognition." Under View.
- P. Miao**, D. Li†, M. Li, W. Liang, Y. Li. "EviComp: Reliable Spatial Evidence Learning from Heterogeneous Annotations for Industrial Visual Inspection" Under View.
- X. Wei*, **P. Miao***, H. Lai†, J. Shi†. "[Learning measurement-critical geometry beyond global segmentation accuracy for reliable 3D anatomical quantification](#)." Under View.

In Preparation

- P. Miao***, X. Wei*, H. Lai†, J. Shi†. "Public-data fine-tuning improves benchmark performance but degrades longitudinal clinical decision-making in implant dentistry in a locally hosted large language model" in Preparation in Preparation

*Equal contribution, †Corresponding author.

RESEARCH EXPERIENCE

Research Assistant / Co-First Author

The Hong Kong Polytechnic University

Mar 2026 — Oct 2026

Hong Kong SAR

- Supervised by [Dr. Dun Li](#) and [Dr. Huan Wang](#)
- Developed **EviComp**, a query-based evidence learning framework for industrial visual inspection under heterogeneous supervision, enabling component, class/binary, union-mask, and image-level inspection records to contribute through a unified evidence interface.
- Designed and constructed **CrystalComp**, a counterfactual compositional wafer benchmark with component-level annotations, spatial evidence masks, and held-out component combinations to study controlled compositional generalization and component-level evidence learning.
- Built **PGC-Agent**, a progressive wafer inspection framework combining uncertainty-gated routing, LoRA-tuned VLM verification, and a physics-grounded reasoning agent for hard mixed-defect cases. Implemented physical evidence tools including 2D-FFT spectra, Radon transforms, radial profiles, KDE density maps, polar views, and structured evidence reports for auditable industrial reasoning.

Research Assistant

Shanghai Ninth People's Hospital, Shanghai Jiao Tong University

May 2026 — Present

Remote, Hong Kong SAR

- Supervised by [Prof. Hongchang Lai](#) and [Prof. Junyu Shi](#).
- Co-developed **Bio-CPD**, detects cliff points (tipping points / critical transitions) along continuous single-cell trajectories such as pseudotime, developmental time, or aging. It scans a sliding window across the trajectory, computes a combined Cliff Point Index (CPI) from Bhattacharyya distance and Wasserstein distance with entropy-based adaptive weighting, identifies peaks, and validates each candidate via logistic-regression AUROC against null distributions — all in a single function call.
- Co-first-authored **SpaGVAE**, supporting method development, benchmarking, manuscript preparation, and public code release for a manuscript currently under review. Implemented entropy-weighted Bhattacharyya/Wasserstein fusion, AUROC-based transition validation, ablation studies, and cross-dataset benchmarking against established change-point methods.
- Co-developed **COMPASS**, a multi-agent framework for clinical research protocol development that transforms implant-related research ideas into structured protocols with evidence retrieval, causal-boundary checks, feasibility review, and expert-facing critique.
- Co-developing **RePlan3D** and **SafePlan3D**, two multimodal AI systems for digital dentistry.

Research Assistant

University College Dublin

Dec 2024 — Jun 2026

Remote, Beijing, China

- Supervised by [Prof. Soumyabrata Dev](#).
- Developed **EmoSense**, an end-to-end pipeline integrating real-time facial-expression recognition at 30 fps with retrieval-augmented temporal emotion summarization and mental-health-oriented report generation. Designed an Adaptive Feature Recalibration Network (AFRN) with dual-pathway attention and progressive feature fusion, achieving 95.39% accuracy on CK+.
- Co-first-authored **SS-VLM**, a spectral-symbolic VLM for auditable facial expression recognition using Fourier-domain visual cues, learned Action Unit priors, retrieval-grounded symbolic reasoning, and constrained report generation. SS-VLM reframes facial expression recognition from a black-box label prediction task into an auditable affective reasoning pipeline.

SELECTED TECHNICAL PROJECTS

Co-Founder

Beijing Shawarma Tech (SAGA AI)

Jun 2025 — Present

Beijing, China

- Built FastAPI backend services for retrieval-augmented generation, AI-authorship detection, streaming LLM responses, and interactive editing assistance.
- Integrated third-party detectors with LLM-driven feature analysis to generate document-level and sentence-level feedback for writing assessment workflows.
- Implemented vector retrieval pipelines with Milvus for embedding storage, semantic search, and evidence-backed feedback generation.

Software Engineer Intern

Tianjin Shengtong Science and Technology Development Co. Ltd.

Jun 2024 — Aug 2024

Tianjin, China

- Developed a B/S three-tier management system for 200+ enterprise users, automating 5+ back-office processes and reducing manual reporting time by 40%.

Database Developer Intern

Haibin People's Hospital of Tianjin Binhai New Area

Jun 2023 — Aug 2023

Tianjin, China

- Optimized hospital database workflows for 50,000+ patient records, reducing redundancy by 25% and improving query speed by 18%.

SKILLS

- Languages:** English (IELTS 6.5; BSc and MSc programmes conducted entirely in English), Mandarin (native proficiency).
- Programming:** Python, MATLAB, JavaScript, SQL, Java.
- Machine Learning:** PyTorch, TensorFlow, VLMs, RAG, multimodal learning, uncertainty-aware routing.
- Computer Vision:** defect recognition, facial expression recognition, spectral analysis, segmentation, heatmap/evidence supervision, 3D/digital-twin research planning.
- Tools:** FastAPI, Milvus, Git, LaTeX, Typst.

AWARDS AND CERTIFICATES

- Honorable Mention, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling, 2023.
- Best Final Year Project (ranked 1/96), BSc Software Engineering, University College Dublin, 2025.
- Deep Learning Specialization, Coursera. [Certificate](#).
- Natural Language Processing Specialization, Coursera. [Certificate](#).